

JOSEPH S. FINBERG

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EDUCATION

Columbia University in The City Of New York

Sep 2021 - May 2024

Bachelor of Arts, Physics & Political Philosophy BA (3.7 GPA)

New York, NY

· **Honors:** Dean's List, ΠΣΑ Political Science Honors Society, American Nuclear Society Scholar.

St Edmund Hall - University of Oxford

October 2022 - June 2023

Year Abroad, Physics, First Class Honours

Oxford, UK

· **Activities :** The Oxford Union Society, SEH Rugby Team , Physics Society

EXPERIENCE

Laurelin Technologies Inc.

August 2024–Present

Co-Founder & Chief Executive Officer

San Francisco, CA

- Founded a nuclear fusion technology company applying advanced AI/ML methods to accelerate reactor design, based on my research experience at Columbia University.
- Lead development of a B2B SaaS platform for fusion-reactor R&D, focusing on plasma stability, turbulence, and optimisation of reactor geometry for net energy gain.
- Built and managed a cross-disciplinary founding team (physics, computer science, finance), overseeing early prototyping and technical roadmap.
- Secured commitments and negotiations toward seed financing (target \$2.2M–\$10M), including discussions with venture capital firms and strategic investors.
- Structured the company's financial model, IP strategy, and investor materials (pitch deck, budgets, SAFE terms) to support fundraising and growth.

Hildred Capital Partners

Aug 2021 – Dec 2024

Quantitative Researcher I

New York, NY

- Developed time-series forecasting models in Python and SQL to predict multi-asset portfolio performance, improving forecast accuracy by 20%.
- Designed and analyzed A/B tests and pilot experiments to evaluate risk-management strategies, applying causal inference methods.
- Created automated ETL pipelines using Python to integrate market and transaction data, reducing data processing time by 30%.
- Built interactive dashboards in Looker to monitor key metrics and experiment results, supporting strategic decisions by senior leadership.
- Conducted ad-hoc statistical analyses and hypothesis tests to uncover trends and inform allocation strategies across trading desks.
- Built and maintained ETL pipelines to ingest and preprocess multi-modal datasets (text, time series, image metadata), cutting data prep time by 30%.
- Designed and executed structured A/B experiments to validate labeling guidelines for financial text classification; applied causal inference to refine annotation criteria.
- Developed interactive dashboards to monitor annotation KPIs (accuracy, throughput, latency), enabling senior leadership to optimise resource allocation.

PRS Financial Group
Quantitative Researcher I

Aug 2019 – May 2021
San Francisco, CA

- Managed end-to-end A/B testing framework for pricing algorithms, performing statistical analysis in Python to optimize revenue by 8%.
- Engineered SQL-based data pipelines to aggregate and transform trade and market data for modeling and dashboarding.
- Developed machine learning models for index performance prediction using scikit-learn, achieving 90% predictive accuracy.
- Built Looker dashboards for real-time monitoring of trading metrics and attribution analysis, enabling faster decision-making.
- Performed causal impact analyses to measure the effect of new strategies, presenting findings to C-suite stakeholders.

Richmond College Preparatory Schools
Summer School Teacher

Jun 2021-Aug 2021
Richmond, CA

- Led a summer school programme, instructing students in grades 6-8 in advanced mathematics.
- Directed an extracurricular astronomy programme for students in grades K–5.

Berkeley City College, Learning Resource Centre
Mathematics Tutor

Aug 2019-Dec 2020
Berkeley, CA

- Worked as a tutor in the mathematics tutoring centre of Berkeley City College
- Helped students many of whom struggled with mathematics improve their abilities.

Eaton Capital Management
Investment Banking Intern

May 2018 – Aug 2018
San Francisco, CA

- Assisted in constructing financial and statistical models in Excel and Python for annual budgeting and forecasting.
- Automated data consolidation scripts in Python to reconcile month-end close data across multiple sources.
- Supported hypothesis-driven analyses to evaluate M&A scenarios and their impact on key performance indicators.

RESEARCH

Fu Foundation School of Engineering and Applied Science
Senior Research Assistant, Columbia Plasma Lab

Oct 2023 - Jun 2024
New York, NY

- Built and executed large-scale gyrokinetic and nonlinear-MHD simulations (Python, MATLAB, Fortran 90) on Columbia's HPC clusters to quantify affine-eigenvalue spectra and growth rates in FRC-like plasmas, directly informing stability limits for next-generation devices.
- Wrote mpi-enabled post-processing scripts to parse terabyte-scale diagnostic output, calculate magnetic-diffusion and eddy-current metrics, and visualise circuit-plasma coupling, cutting analysis cycle-time by 60 %.
- Communicated findings through rigorously documented LaTeX reports and technical presentations, aligning experimentalists and engineers on design iterations under tight development schedules.
- Labeled and classified terabyte-scale MHD simulation outputs (text logs, diagnostic images) using MPI-enabled postprocessing scripts; documented all procedures in LaTeX reports.
- Analyzed root causes of annotation discrepancies, proposed guideline enhancements, and trained six new annotators on SOPs.

- Integrated SPICE-based coil models with annotated data to generate prototype design reviews and DOE milestones.

Columbia University, Marka Lab

Senior Research Assistant

Nov 2021 - May 2022

New York, NY

- Developed an algorithm to analyse gravitational wave parameters using a variational autoencoder.
- Worked alongside a hardware team that developed an ASIC solution for analysing gravitational waves.

PUBLIC SERVICE

The Oxford Union Society

Special Advisor To The President for American and Transatlantic Affairs

Nov 2022 - June 2023

Oxford UK

- Functioned Essentially as a Junior producer for the oldest debating society and political event society in the world.
- Designed speaking, debate, and panel events for our members.
- Helped Secure events for Senator Bernie Sanders, Journalist Robert Evans, and Steve Coll & Secured additional funding for The Union from VIP Donors

PUBLICATIONS

9. "Analysing the Language of the Freedom Caucus" (2021)
8. "Philosophical Implications of Conformal Cyclic Cosmology" (2021)
7. 'Dasein und Unsinn A Critique of Heidegger's Non-Materialist Temporal Epistemology', Logos, Cornell University Logos Journal, online (2023)
6. 'Freedom from Religion, not OF How The Dobbs decision is a violation of The Establishment Clause.' Columbia Undergraduate Law Review, Summer (2023)
5. "China's Nuclear Tellurocracy" (2023)
4. "Tractatus Temporalis Secundus" (2023)
3. A Unified Hamiltonian Formulation for Energy Loss, Entropy Evolution, and Fusion Performance in Plasmas, <https://arxiv.org/abs/2412.07725> (10 December, 2024)(an improvement upon my work at CU Plasma Lab)
2. Remarks on the Fundamentals of Time oder die Zeitdeämmerung, <https://arxiv.org/abs/2508.14312>(21 Aug 2025)
1. Collisionless Relaxation and Universal Equilibria in Magnetised Plasmas: A Lynden-Bell Theory for the Vlasov-Maxwell System (Forthcoming)

REFERENCES

Name	Relationship	Phone	Email
Debra Solomon	Personal	+15165107637	debra@spectrumroadmap.com
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